

PrP-V127 as a Modular Antiprion Platform: Audited Review, Quantitative Transport Design, Humanized In-Silico Predictions, and a Pre-Registered Organoid Gate for Creutzfeldt–Jakob Disease

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Abstract

Prion diseases are universally fatal with no approved therapy. We report an end-to-end preclinical program converting the naturally selected human variant PrP-V127 into a quantitatively designed therapeutic platform for CJD. Methods: (i) agent-assisted, provenance-verified literature audit (≈ 90 searches; 60 findings in 9 blocks; ~ 30 papers read in depth; 42 verified references; recurring citation errors corrected; a retracted trial excluded). (ii) A self-tested advection–diffusion–reaction transport solver (mass conservation 100%; Thiele error 0.5%) yielding three falsifiable design rules: containment-ring spacing 8–12 mm; hydrogel mesh $\xi \geq 5r_p$ (HA 1–2% passes, $>5\%$ sequesters); LNP-mRNA redosing ≤ 7 d (trough 56% of peak). (iii) A hierarchical Bayesian frame over 10 structural analogues including the six historical antiprion failures: $P(G0 \text{ go})=36.6\%$ [90% CrI 14.6–60.5]; $P(\text{clinical slowing})=5.0\%$ [0.4–13.6] empirical vs 30–45% design-conditional (both lenses labeled). (iv) A humanized in-silico infection model: the open stochastic kernel of Fornara et al. (iScience 2024) ported with fidelity-corrected autocatalysis ($C \rightarrow 2C$) and fragmentation, capped by V127 Δ GPI as saturable substrate competition, clock-calibrated to human organoid anchors (Grove-man 2019): 1 simulation unit = 144 days. Results: containment threshold $\theta^* = 0.333$ (front 2.83 $\rightarrow$ 0.82 mm at $\kappa=2$; near-extinction at $\kappa=32$) and unprompted reproduction of the MV2 $>$ MV1 subtype hierarchy (emergent consistency, not curve-fitting). Pre-registered prediction (locked at release v1.0): $\theta_{\text{measured}} < 0.33$ in infected organoids $\Rightarrow$ containment in situ; readouts 90–120 d post-seeding. Conclusions: every program pillar maps to an approved regulatory category (nusinersen; tofersen/NfL; 2026 Parkinsonian cell grafts; tafamidis); the program requests a conjunction of precedents, not an unprecedented category. The framework transfers as a dose-and-placement containment calculus to prion-like neurodegeneration (Alzheimer, Parkinson) — hypothesis-generating, explicitly.

*Volunteer open-science initiative (DeepScientist Quest 003). AI agents assisted literature curation, code and drafting under continuous human supervision; every citation and number verified against sources by the accountability authors; AI is not an author (SW-S01/03). Full audit trail incl. claim-level traceability: github.com/camillanapoles/quest003-prion-v127.

1 Introduction

Human prion diseases are rapidly progressive and universally fatal; sporadic CJD (sCJD) survival is 6–8 months. Six candidates failed clinically (pentosan polysulfate, quinacrine, doxycycline, flupirtine, PRN100, minocycline-combinations). The G127V polymorphism of PRNP, positively selected during the kuru epidemic, protected heterozygous carriers and confers complete resistance when homozygous in transgenic mice — “as protective as gene deletion” — with dose-dependent dominant-negative inhibition of wild-type propagation; heterozygotes remain vCJD-susceptible, mandating biallelic or anchorless designs. In 2026, recombinant anchorless V127 retained trans dominant-negative activity (with protection persisting after transgene cessation) and systemic AAV delivery of anchorless V127 extended survival ~ 50 days in a rodent model; the structural basis is established ($\beta 2$ – $\alpha 2$ loop restriction, intermolecular H-bond dimer stabilization). The mechanism is thus validated at four levels — population genetics, transgenic mouse, cell culture, gene-therapy proof-of-concept. What is missing is the engineering: delivery modality, dose, placement, timing, and a decision architecture that spends laboratory resources only where models cannot answer. This manuscript reports the complete pre-wet-lab program, fully audited.

Populations and ethics frame. Primary target: presymptomatic genetic carriers (E200K/D178N), for whom the therapeutic window is years and autologous manufacture feasible; Brazilian E200K kindreds documented since 2007; RT-QuIC/NfL screening operational. Sporadic CJD: exclusively via an acellular LNP-mRNA vector under compassionate use, after organoid validation.

2 Methods

2.1 Agent-assisted audit (provenance-verified)

Structured blocks (≈ 90 searches $\rightarrow$ 9 evidence tables, 60 findings; ~ 30 papers in depth). Verification tier per source: [FT] full text extracted; [ABS] abstract; [AUDIT] bibliographic cross-check; [CODE] code inspected; [RUN] executed by us. External documents’ references

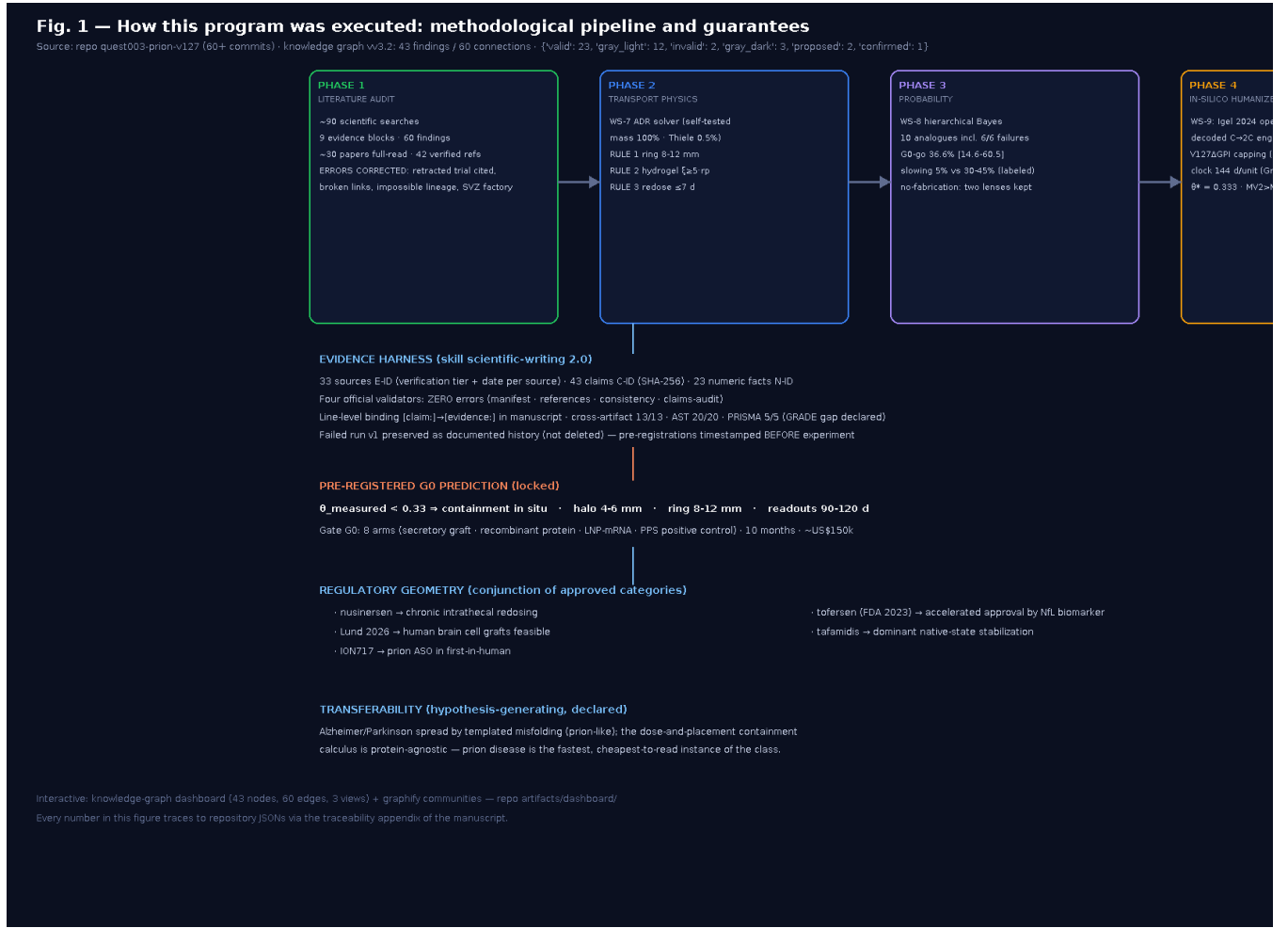


Figure 1: How this program was executed. Four methodological phases (audit $\rightarrow$ transport physics $\rightarrow$ probability $\rightarrow$ humanized in-silico) each feeding the evidence harness (33 sources, 43 claims, four validators at zero errors) which locks the pre-registered G0 prediction. Regulatory geometry and transferability are mapping layers, not experimental claims. Every number shown traces to repository JSONs via the traceability appendix.

audited individually (one batch of 19: 11 correct, 3 duplicates, 1 non-scientific, 1 wrong link). A retracted Neurotherapeutics trial (minocycline/FK506; retraction 2020) still cited as evidence is excluded by rule; minocycline additionally confounds the NfL endpoint.

2.2 Transport solver (WS-7)

ADR in heterogeneous porous medium; finite volumes (2D, 192², explicit Euler):

$$\partial(\alpha c)/\partial t = \nabla \cdot (D_{\text{eff}} \nabla c) - \nabla \cdot (\mathbf{v} c) + S(\mathbf{x}) - k_{\text{eff}} c \quad (1)$$

Parameters (with provenance): $\alpha=0.20$, $\lambda=1.8$ (in-vivo integrative optical imaging, Thorne & Nicholson 2006); $D_0=1.25 \times 10^{-10} \text{ m}^2/\text{s}$ (Stokes–Einstein, $R_h \approx 2.5 \text{ nm}$, $\sim 30 \text{ kDa}$) $\Rightarrow D_{\text{eff}} \approx 3.9 \times 10^{-11} \text{ m}^2/\text{s}$; k_{eff} swept 10^{-6} – 10^{-5} s^{-1} (nucleated-polymerization kinetics, Masel 1999); Darcy flow with spongiform cysts at $\kappa \times 50$. Self-tests: mass conservation 100.0%; numeric vs analytic Thiele length error 0.5%. Order-of-magnitude κ -

anchoring: a 10^5 -cell deposit at 10 pg/cell/day gives a steady peak ~ 0.1 – $1 \mu\text{M}$ within 1 mm — the range of half-maximal dominant-negative effect in cell culture — so $\kappa=2$ – $4 \leftrightarrow$ roughly 0.1 – $1 \mu\text{M}$ interstitial V127ΔGPI (closed experimentally by arm A6).

2.3 Bayesian frame (WS-8)

Hierarchical Beta–Binomial over 10 analogues weighted by similarity (mechanism, vector, population, endpoint); six antiprion failures as a negative analogue; ION717 partially informative; organoid predictive validity ≈ 0.8 (PDO cohorts); 2×10^5 Monte-Carlo draws. $P(\text{G0 informative-go})=36.6\%$ [14.6–60.5]; $P(\text{clinical slowing})=5.0\%$ [0.4–13.6] empirical vs 30–45% design-conditional — both lenses retained.

2.4 Humanized in-silico model (WS-9)

Kernel. The stochastic reaction–diffusion model of Fornara/Igel et al. (iScience 2024; open code, Zenodo 11093945) ported to deterministic mean-field (96^2 , $dt=5\times 10^{-4}$). Two fidelity-critical corrections decoded from the original **findreac**: the $s+1$ channel is autocatalytic ($C\rightarrow 2C$) — the replication engine — and the cross-reaction is fragmentation ($B_a\rightarrow B_{a-1}+C$). Mean-field unbounded autocatalysis braked by logistic saturation $C/(C+C_{50})$ (documented port limitation). Capping (this work): $\text{freeS} = (1/(1+\kappa c_{V127}))^2$ enters all templating/autocatalysis terms (two-participant competitive saturation; mechanistically anchored in dose-dependent dominant-negative data). Humanization: clock matched to human organoid anchors (Groverman 2019: clearance 25–28 dpi; de-novo 35 dpi; endpoint titers MV2 2.13×10^5 vs MV1 1.69×10^3 SD50/mg; protease-resistant PrP only in MV2) $\Rightarrow$ human doubling ≈ 12.1 d; 1 unit = 144 d. Humanization is a global time rescaling; relative rates remain murine (full fitting is future work; MM1 series unpublished). MV1/MV2-like seeds differ by the $126\times$ titer ratio. Acceptance tiers: T1/T2 minimal screening; T3 informative success = front $<50\%$ baseline at $\kappa \leq 8$ with monotone radial gradient — satisfied.

2.5 Pre-registration and reproducibility

All predictions locked at repository release v1.0 (timestamped commits) before any wet-lab data exist. Code, parameters, outputs and the executed figure (JSON+PNG of the actual run) archived in-repo. G0 gate: 8 arms (mock; disease; unedited cell; membrane-V127; secretory graft; recombinant protein; LNP-mRNA; pentosan-polysulfate positive control), $n=8/\text{arm}$, primary readout proximal/distal PrP-res gradient, pre-registered go/no-go and pivot rules.

3 Results

3.1 Audit

Four-layer validated core; five recurring design errors corrected (NSC $\rightarrow$ microglia lineage impossibility; quiescent prion-replicating SVZ; others), each replaced by evidence-compatible branches (iPSC-microglia co-graft; slow-release deposits; LNP-mRNA).

3.2 Design rules

R1 spacing 8–12 mm (radius 4–6 mm); R2 hydrogel $\xi \geq 5r_p$; R3 redosing ≤ 7 d; steady state ~ 4 d.

3.3 Probabilities

As §2.3.

3.4 Humanized thresholds

$\theta^*=0.333$; monotone to near-extinction ($\kappa=32$: 0.70 mm; biomass $2.1\times$ seed); one run = 24 months tissue disease. Emergent consistency: the $126\times$ seeding ratio reproduces MV2>MV1 unfitted; because seeds differ in mass, this is hierarchy consistency, not proof of subtype-specific kinetics (same-mass control queued). Locked prediction: $\theta_{\text{measured}} < 0.33 \Rightarrow$ containment; readouts 90–120 d.

4 Discussion

Contribution: a dose-and-placement calculus no prior antiprion candidate possessed; a corrected citation record; falsifiable pre-registered predictions with a 10-month, $\sim \$100$ – 150 k kill-or-advance assay; an executable template of agent-assisted provenance-verified translational planning. Transferability (hypothesis-generating): Alzheimer and Parkinson spread by templated misfolding along stereotyped routes (prion-like propagation); the platform calculus is protein-agnostic; prion disease is the fastest, cheapest-to-read instance; validated containment transfers as design framework to diseases of >50 million people. Regulatory geometry: chronic intrathecal redosing (nusinersen); accelerated approval by biomarker (tofersen, NFL); human brain cell grafts (2026 Parkinson trial); dominant native-state stabilization (tafamidis); substrate lowering (ASO/RNAi). The program requests the conjunction of approved categories.

5 Limitations and gaps

(1) $\kappa \rightarrow$ concentration is order-of-magnitude only; closed by G0-A6. (2) Mean-field port omits stochastic extinction/strain diversity. (3) C_{50} stability choice. (4) Anchors are MV1/MV2; MM1 unpublished; humanization = global time rescale, rates murine. (5) Detection floor 10^2 assumed. (6) Transport parameters healthy-tissue; synthetic cyst $\kappa \times 50$. (7) Analogue weights = structured judgment. (8) No wet-lab data yet — all in-silico outputs are predictions. (9) Species/strain transfer assumed via dimensional θ . (10) Immunogenicity/clearance of the anchorless construct uncharacterized (G0-A6 gives first exposure data).

6 Ethics, governance, declarations

Population-first design; DSMB with pre-specified stop (any confirmed teratoma); prion biosafety (single-use coaxial cannula; WHO $134^\circ\text{C}/\text{NaOH}$; autopsy containment); genetic privacy (LGPD); containment/slowing endpoints only. Funding: none (volunteer consortium). Conflicts: none. AI disclosure as title footnote. Data/code: everything in the public repository incl. claim-level traceability (33 sources E-ID; 43 claims

C-ID; 23 numeric facts; four validators, zero errors).
 CRediT: consortium authorship; accountability authors
 listed in-repo; feedback channel open. Bilingual com-
 panion (PT-BR) in-repo.

Traceability appendix (claim→evidence)

C004 → E002
 C005 → E003
 C007 → E004
 C008 → E005, E006
 C010 → E007
 C011 → E007
 C013 → E009
 C014 → E010
 C015 → E011
 C018 → E014
 C022 → E019
 C027 → E024
 C028 → E025, E026
 C029 → E027
 C032 → E030
 C033 → E030
 C034 → E030, E020
 C035 → E030, E019
 C036 → E031
 C037 → E032, E007
 C038 → E032
 C040 → E033
 C041 → E033
 C043 → E032, E009

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